AI Application R&D Engineer · BenQ Materials

Shan-Cheng Chang 張善程

LLM Agent Engineering · Knowledge Graph · Document AI

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— CORE SKILLS

Agent Engineering

Agent Harness · Function Calling · MCP · Skills / Tools · Context Management · Sandboxing · RBAC / Audit Log

Evaluation & Validation

Golden Set / Held-out · Code Validators · Regression Baseline · Failure Analysis

LLM & Retrieval

Gemini API · Azure OpenAI · Prompt Engineering · RAG · Knowledge Graph (Neo4j) · pgvector

Document AI / ML / CV

Azure Document Intelligence · OCR + LLM Extraction · YOLOv8 · PyTorch

Backend & Web

Python · FastAPI · Node.js · React · TypeScript · PostgreSQL · SQL Server

DevOps & Tools

Docker · GCP · Azure · Git · Claude Code

Security

AES-256-GCM · JWT · AD / Entra ID · ISO 27001

— EDUCATION

National Taiwan University
M.S. Biomechatronics Eng. (AIoT / CV) · 2021 – 2023

National Taipei University of Technology
B.S. Mechanical Engineering · 2017 – 2021

— LANGUAGES

Mandarin Chinese — Native
English — Professional (TOEIC 790)
German — A2
Company Digital Talent L3 (Expert)
Military service completed

AI application engineer who designs, verifies and maintains enterprise AI systems in manufacturing: lead developer of BenQ Materials' on-prem AI data-analysis platform and Claude Code–style agent harness, in daily production use, after building the company's document-AI stack from zero. Every system ships with a labelled test set or regression check.

4.15M NTD	6	~1M NTD	1,500+ h
AUTOML PURCHASE AVOIDED / 2 YRS	PRODUCTION-LINE FLOWS ON PLATFORM	TANGIBLE BENEFIT / YR (3 OCR)	SAVED / YR (3 OCR SYSTEMS)

— EXPERIENCE

AI Application R&D Engineer

Dec 2024 – Present

BenQ Materials Corp. (明基材料) · Digital Technology Center, Intelligent Application Dept., Intelligent Application Development Sec. · Taoyuan, Taiwan

- **Enterprise Brain → AI data-analysis platform** (system architect & lead developer) — live on company servers and used daily by the BML polarizer line, replacing a purchased AutoML package (**4.15M NTD / 2 yrs**); 6 production-line report flows cut over with zero downtime. Claude Code–style harness on FastAPI + Gemini function calling + MCP, 10 Docker services, data stays on the intranet.
  - **Pluggable architecture** — designed the platform so analysis methods and ML models built by teammates plug in as skills, with no platform changes; Excel uploads, the API and 6 scheduled flows all call the same skill. Skills use Claude Code's format, so teammates build and test in Claude Code and deploy unchanged. Code computes the statistics and the LLM only writes the interpretation, so a failed LLM call never blocks the numbers.
  - **Migration & verification** — 6 n8n workflows (~550 nodes) → 1 skill + 7 code flows, checked against the old output (record counts, live replays, byte-identical PDF reports) before cutover.
  - **Knowledge loop** — engineers annotate analysis results with their experience → AI checks them against existing knowledge → a reviewer approves → knowledge graph. Chat Q&A opens once the knowledge base is mature.
  - **Reliability & safety** — long conversations are auto-summarized to stay within the model's limit; failed tools are retried and the error goes back to the model to try another way; at most 10 tool calls per question to prevent loops. Each role sees only its allowed tools, company rules (e.g. never read secret files) are enforced in code rather than prompt wording, code runs in a resource-limited sandbox, and every action is written to a tamper-proof audit log.
- **Document-AI applications (OCR + LLM)** — built the company's platform from zero (Azure Document Intelligence + Azure OpenAI, Python/FastAPI, React, SQL Server) and its reusable OCR → LLM → validation → feedback pipeline; each system scored on a hand-labelled test set before go-live.
  - **Quotation OCR** — validity-date accuracy **51% → 100%** (29-document regression set; quote date 86%); 30 s → 10 s per document; 758+ documents processed.
  - **Taiwan & China invoice OCR** — **31/32** (5 key fields all correct, incl. handwritten) and **31/31** (10 fields); saves 1,447 h/yr; live at 3 sites.
  - **Health-check OCR platform** — **97.4%** field-level accuracy (42 reports × 46 fields), 216 reports processed; AES-256-GCM, AD + JWT, RBAC and audit logging to ISO 27001.
  - **APO-OCR prompt-optimization engine** — designed the eval-driven loop: 32-invoice golden set (24 train / 8 held-out, 3 leakage paths removed), exact-match code validator (LLM proposes, code scores), budget/loop/failure breakers; mentored an intern to ship it — **8/8** held-out, tying the hand-tuned prompt, for $1.44.
- **Google Cloud Platform — supplier questionnaire management system** (live) — React 19 + Cloud Functions + Firestore; 52 APIs, Entra ID login, built solo in 4.5 months, handed over with GCP training.
- **ESG carbon & energy platform refactor — 44K → 6K lines**, 5 → 1 hosts, validated web upload instead of a personal-PC import.

Shift Supervisor (值班股長)

Jan 2024 – Nov 2024

China Steel Corporation (中國鋼鐵) · Cold-Rolling Mill · Kaohsiung, Taiwan

- **Ran shift operations** on the cold-rolled coil line, keeping production continuous at **95% yield**.
- **Data-driven anomaly analysis** — process parameters (rolling force, tension, speed, temperature) plus quality data; SPC, trend and correlation analysis to find root causes of coil defects and downtime.
- **Anomaly & defect reports** (Python / Excel) for daily production meetings — the domain base for the later production-line AI platform.

— RESEARCH & SELECTED PROJECTS

AIoT Pest Monitoring & Bee Pollination (M.S. thesis, NTU)

2021 – 2023

Lead developer of greenhouse AIoT systems: YOLOv8 + StrongSORT bee tracking (*99.5% mAP*), Mask R-CNN pollen segmentation, deep-learning pest counting (*95.8%*), kriging-based climate control.

PyTorch · YOLOv8 · StrongSORT · Mask R-CNN · IoT · Kriging

Universal OCR Platform — no-code OCR builder (side project)

2025 – 2026

Define fields and prompts, run a 7-step OCR → LLM pipeline, deploy as a REST API; three cloud variants (FastAPI · React 19 · Supabase · Cloud Run · Docker).

— PUBLICATIONS & PATENT

Chou, C.-Y., **Chang, S.-C.**, et al. (2023). Development of AIoT system for facility asparagus cultivation. *Computers and Electronics in Agriculture*, 206:107665. *(co-first author with advisor)*

Chang, S.-C., et al. (2022). AIoT based pest counting system for asparagus cultivation. *Proc. 10th ISMAB.*

Taiwan Invention Patent TW I842416 B — Crop Harvester, 2024.